ROBOTIC ARM STRUCTURAL SIMULATION

ROBOTIC ARM AFTER CAD CLEAN

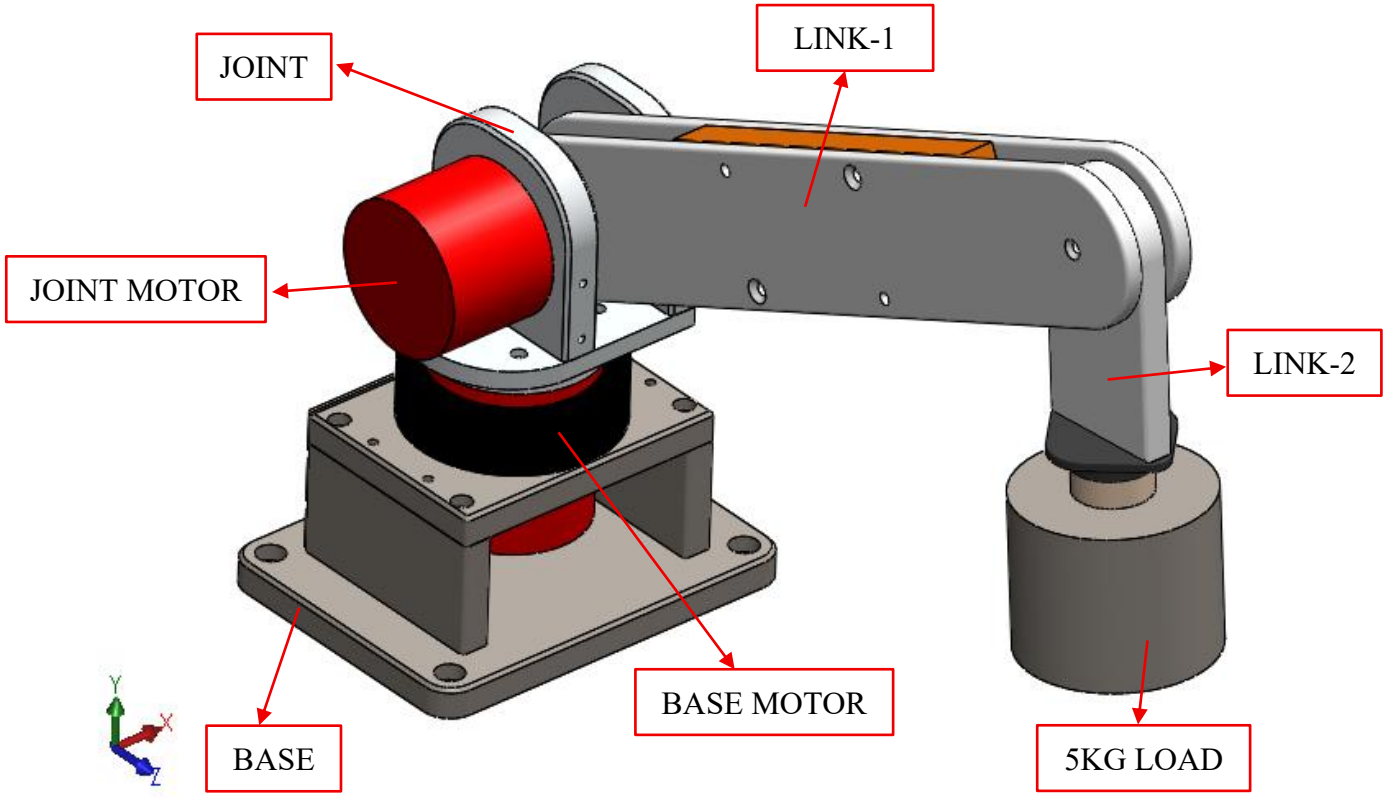


Figure-1: Robotic Arm After CAD Clean Up

MODEL DESCRIPTION FOR STRUCTURAL SIMULATION

- **Base Assembly**
  The base is modeled as a single rigid body integrated with the motor housing.

- **Joint Assembly**
  The joint is modeled as a single body consisting of:
  - Joint motor
  - Bearing housing
  - Pin shaft

The rotor motion is defined about the **Y-axis** of the central coordinate system.

- **Link-1 Assembly**
  Link-1 is modeled as a single rigid body, including:
  - Arm-1
  - Arm-2
  - Cross member
  - Bearing holding pin shaft

- **Link-2 and End Effector**
  Link-2 and the end-effector electromagnet are combined and modeled as a single body (Body-4).
  A payload of **5 kg** is applied at the end effector.

- **Model Simplifications**
  - Bolts, screws, and bearings are removed from the geometry.
  - Small features are suppressed to reduce computational cost.
  - Idealized contacts are defined where required.

These simplifications are made to improve mesh quality and reduce simulation time without significantly affecting global structural behavior.

Software Used: Ansys 2023 R1

SIMULATION PARAMETER:

Geometry:

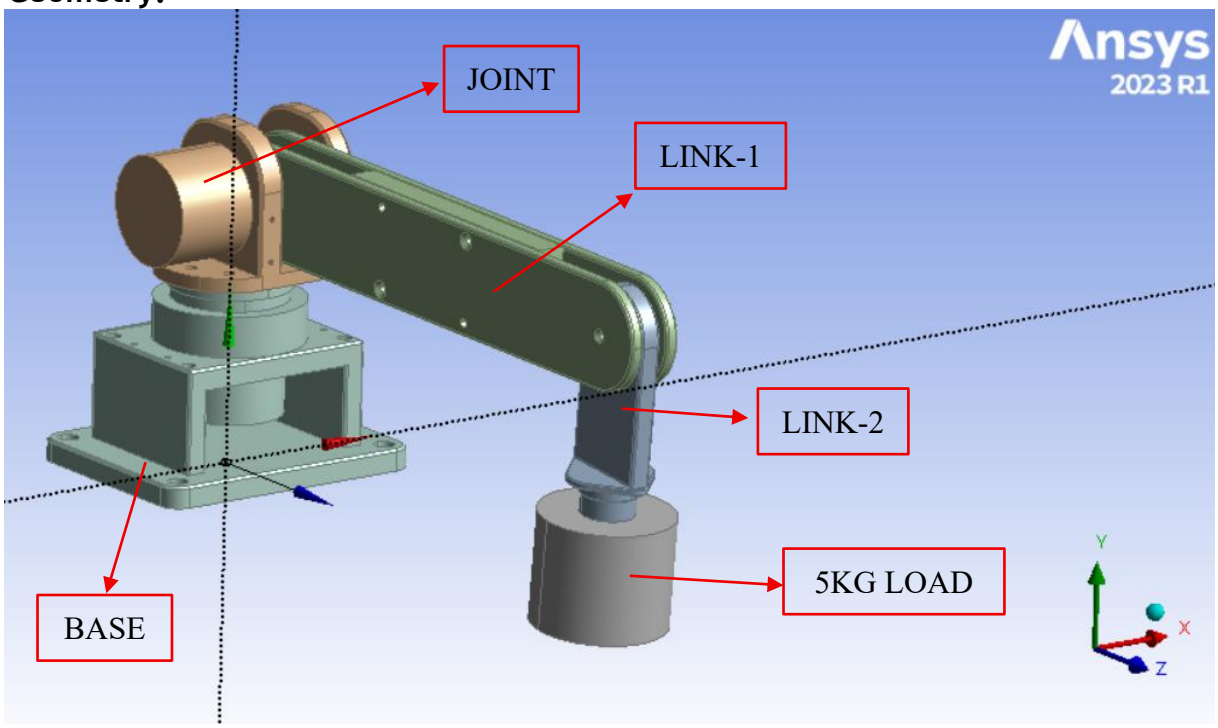


Figure-2: Robotic Arm with 5 bodies

Material:

Base	Mild Steel – AISI 1010	▪ Density = 7872 kg/m3 ▪ Tensile & Compressive yield Strength = 250 MPa ▪ Tensile Ultimate Strength = 460 MPa ▪ Youngs Modulus = 200 GPa ▪ Bulk Module = 166.6 GPa ▪ Poisson's Ratio = 0.3
Joint	Mild Steel – AISI 1010	
5kg Load	Mild Steel – AISI 1010	
Link-1	Aluminium Alloy 6061	▪ Density = 2713 kg/m3 ▪ Tensile & Compressive yield Strength = 259.2 MPa
Link-2	Aluminium Alloy 6061	

		▪ Tensile Ultimate Strength = 313.1 MPa ▪ Youngs Modulus = 68.9 GPa ▪ Bulk Module = 58 GPa ▪ Poisson's Ratio = 0.33

Meshing

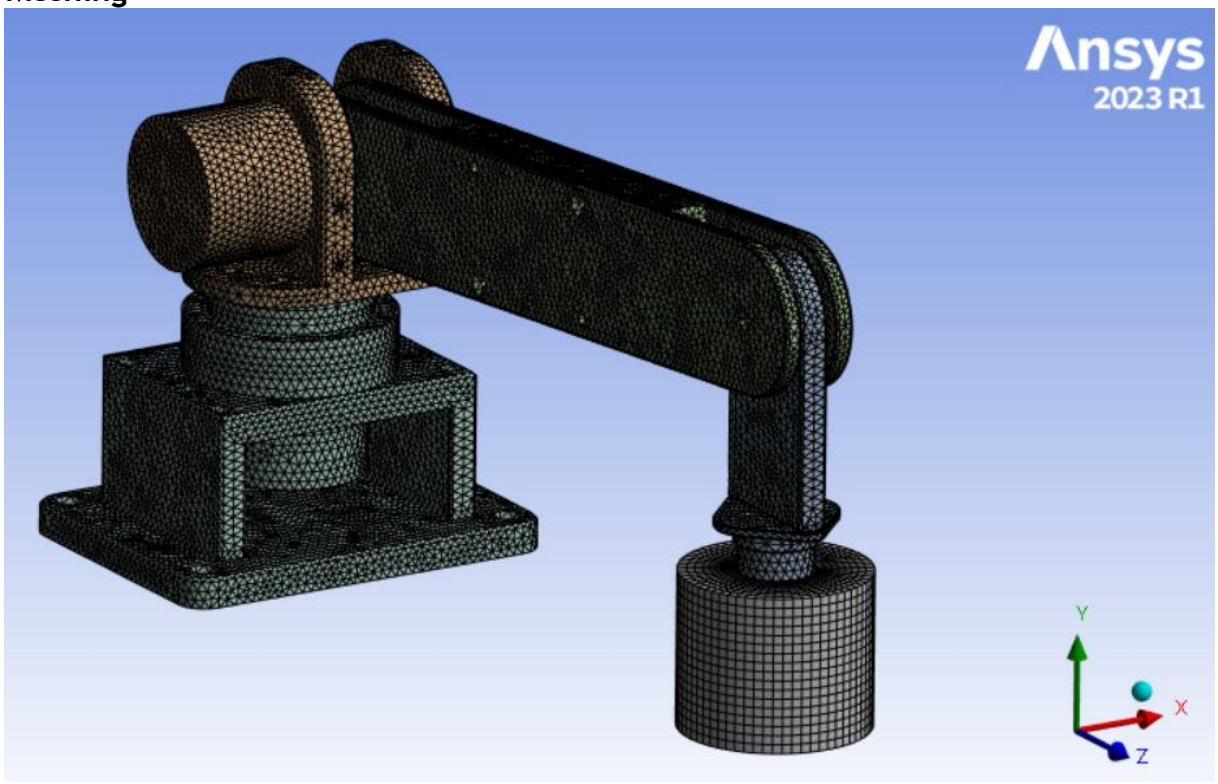


Figure-3: Body mesh with 5mm element size

STATIC STRUCTURAL SIMULATION

Load & Constrains:

- Base Body: Fixed
- Earth Gravity: -ve Y axis
- Load: 5kg Load self-weight

At Pick Up Position

Total Deformation

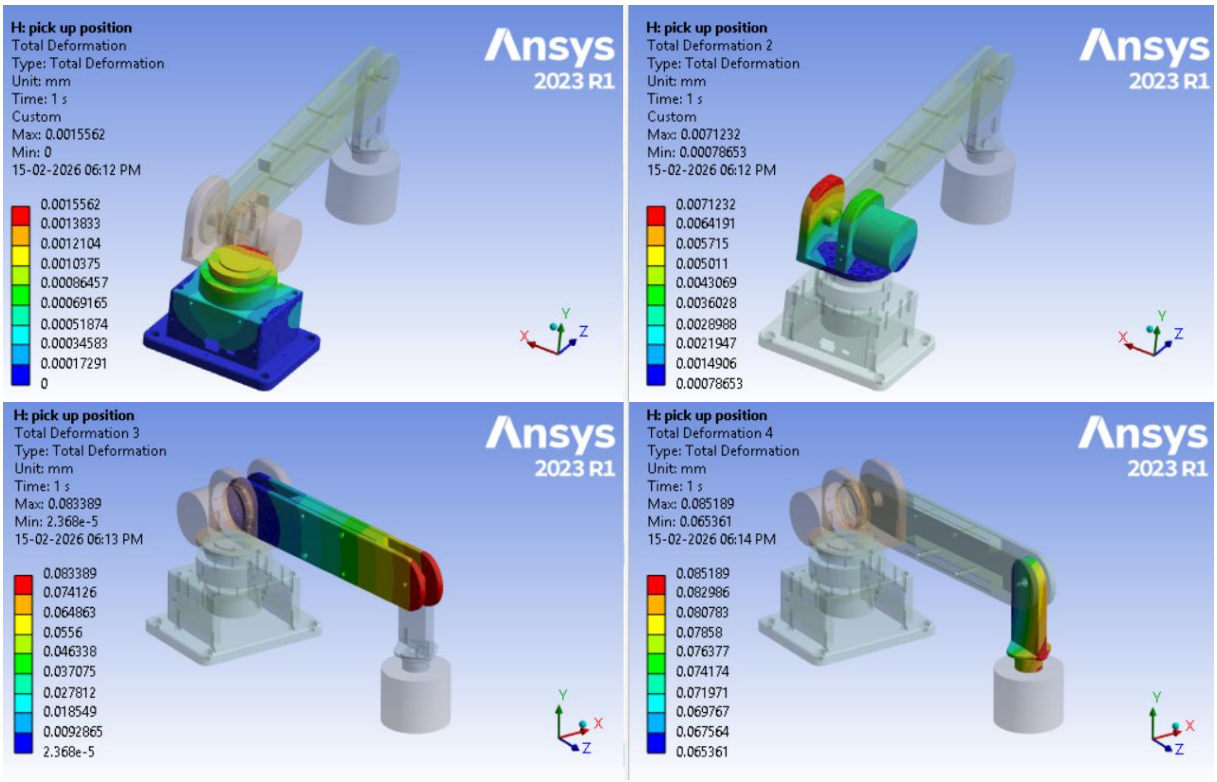


Figure-4: Total Deformation of 4 Bodies at object pick up position

Equivalent Stress

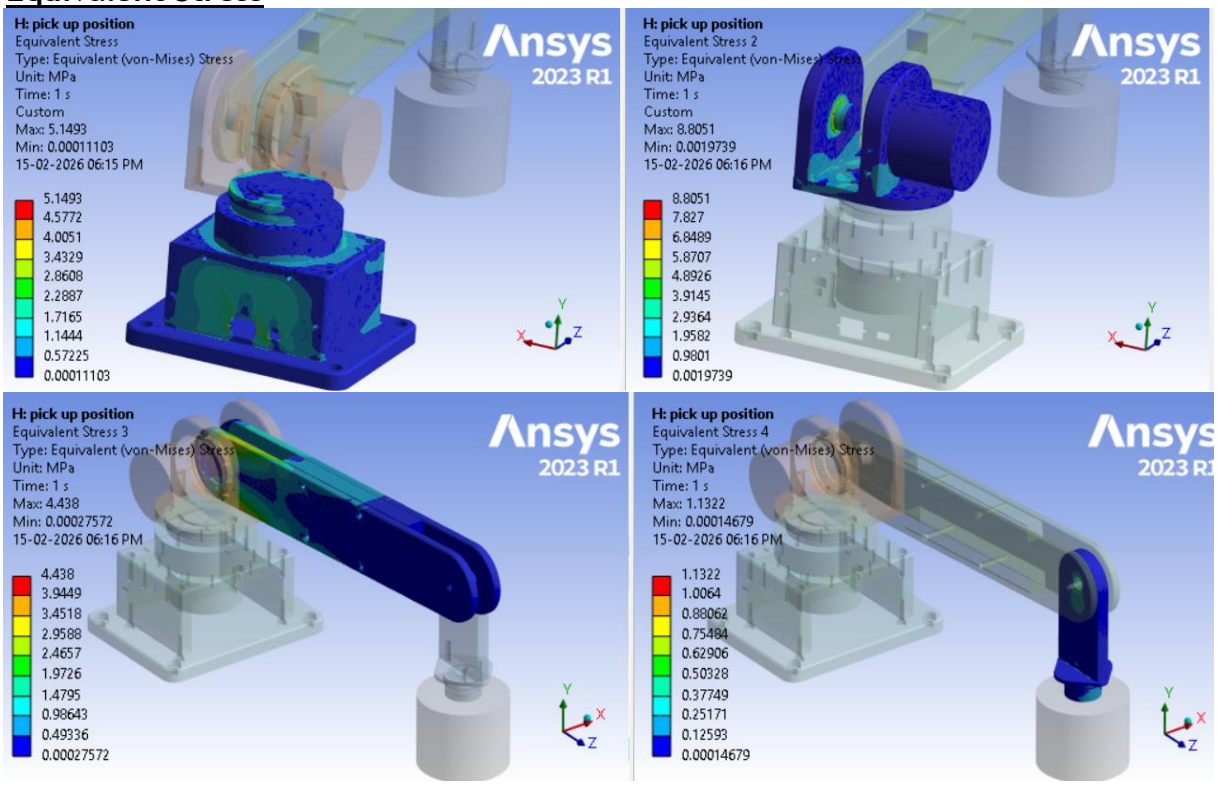


Figure-5: Total stress acted on 4 bodies

Factor of Safety

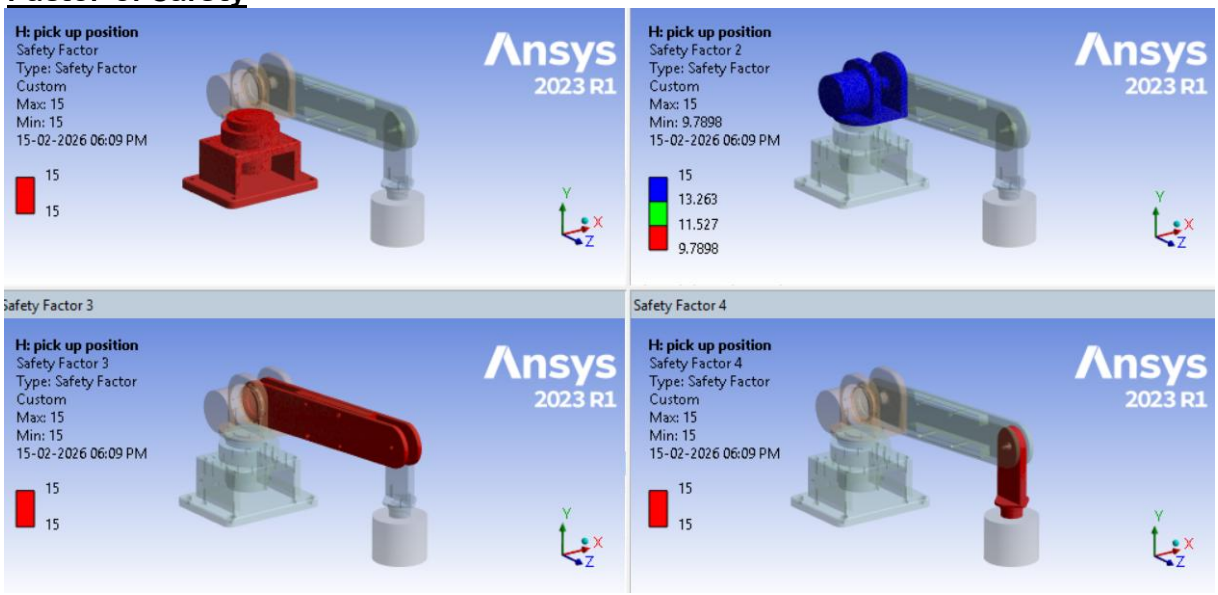


Figure-6: FOS of 4 bodies

Simulation at Holding Position

Total Deformation

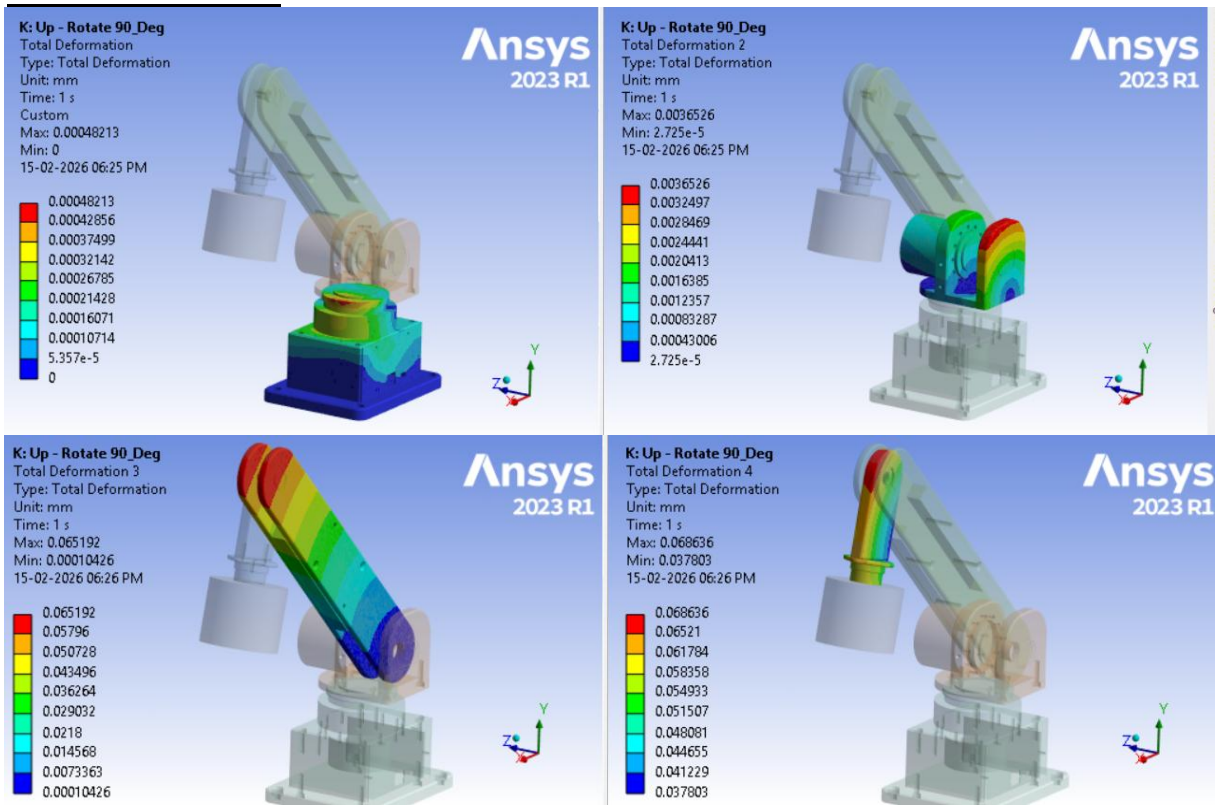


Figure-7: Total Deformation of 4 Bodies at object holding position

Factor of Safety

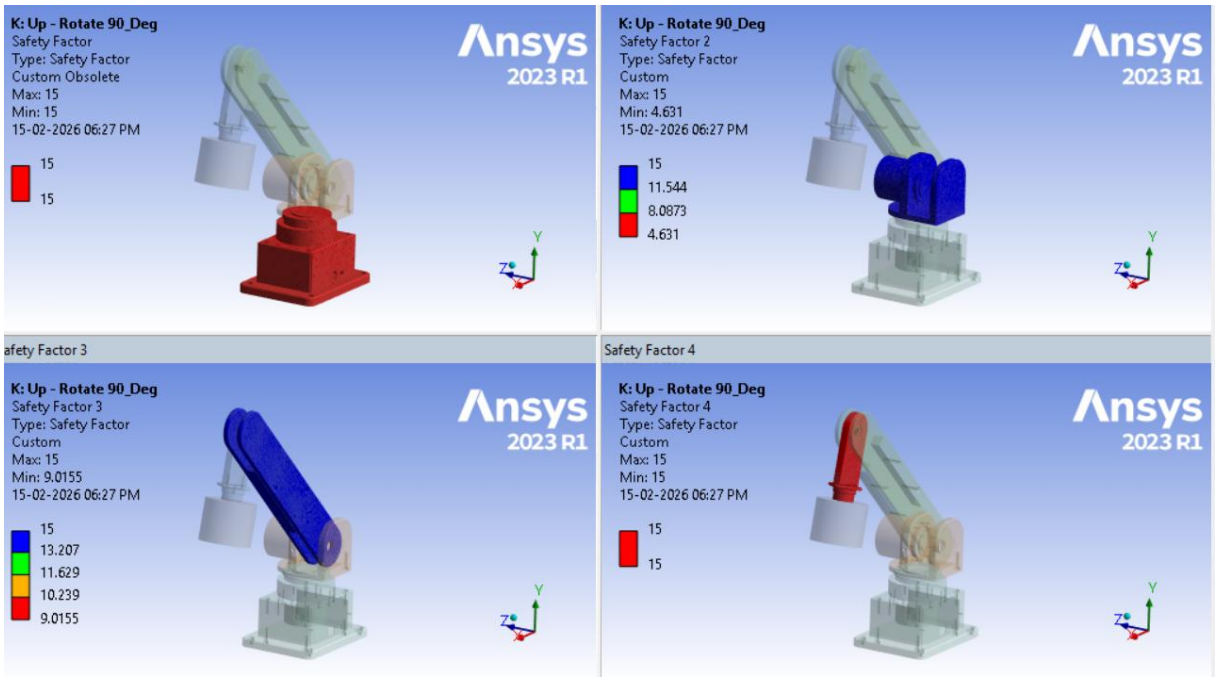


Figure-8: FOS of 4 bodies

TRANSIENT STRUCTURAL SIMULATION:

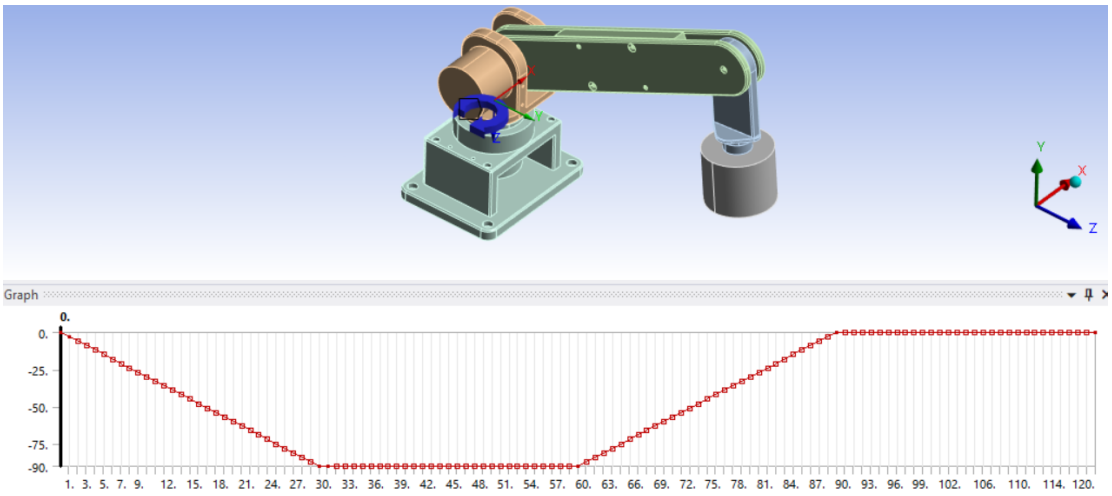
A transient structural analysis was performed to evaluate the dynamic response of the robotic arm during a pick-and-place operation. Prescribed rotational motion was applied to the following components:

- Joint (Base Motor Shaft): Rotates from 0° to 90°
- Link-1: Rotates from 0° to 25°
- Link-2: Rotates from 0° to 15°
- The applied motion sequence enables the arm to:
- Pick the object from the origin.
- Rotate 90° and place the object at the target location.
- Return by rotating back.
- Pick the object again.
- Rotate 90° back to the origin and drop the object.

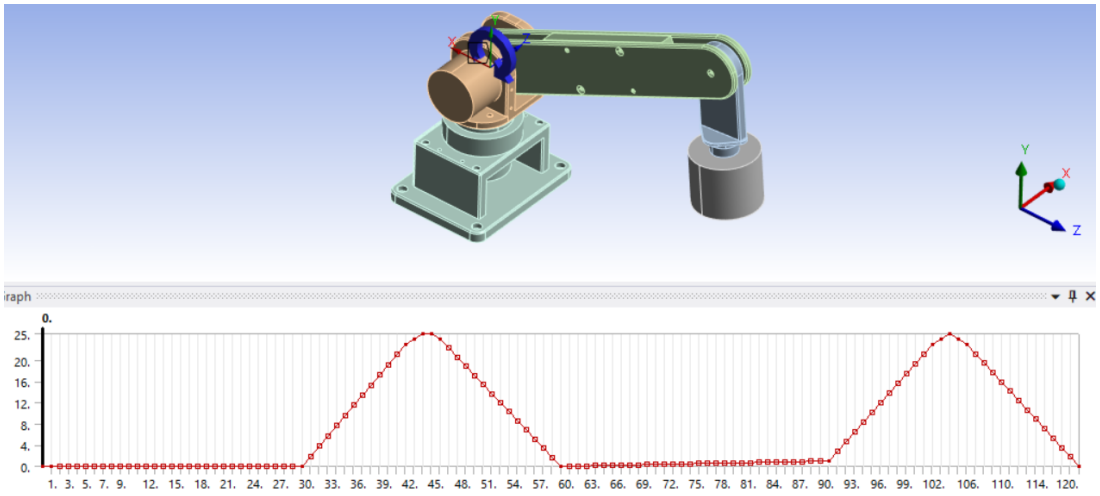
The total operation cycle duration is 120 seconds (tentative). The motion is applied as a time-dependent rotational displacement. Smooth ramping is assumed to avoid unrealistic acceleration spikes.

The base is fixed throughout the simulation. A 5 kg payload is applied at the end effector during the pick phase. Gravity is included in the analysis.

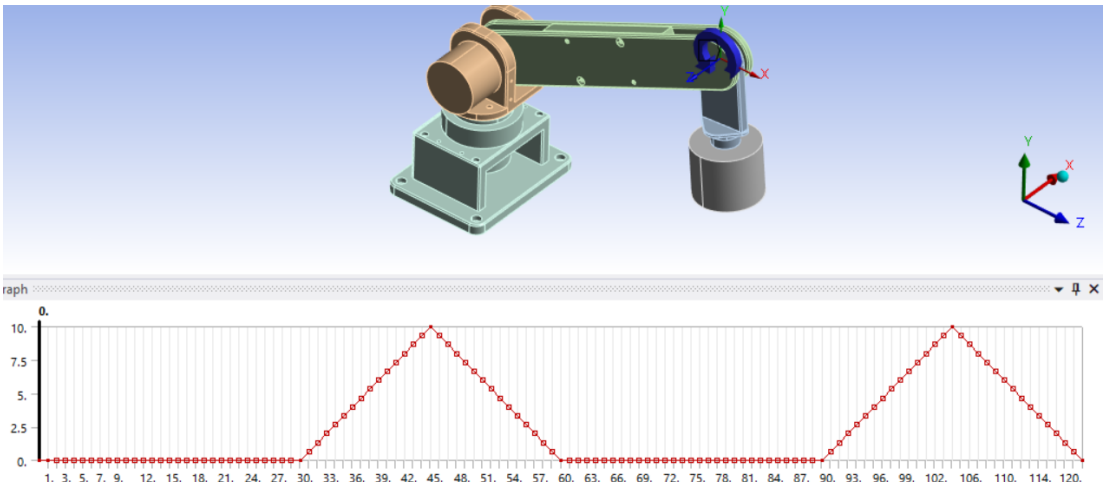
Joint Rotation:



Link-1 Rotation:



Link-2 Rotation:



Result:

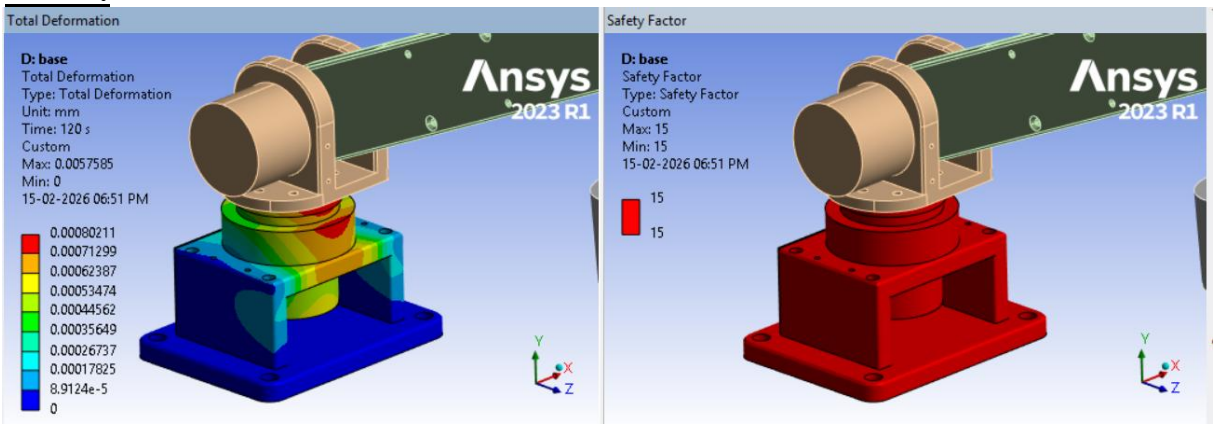


Figure-9: Total Deformation & FOS of base body at motion

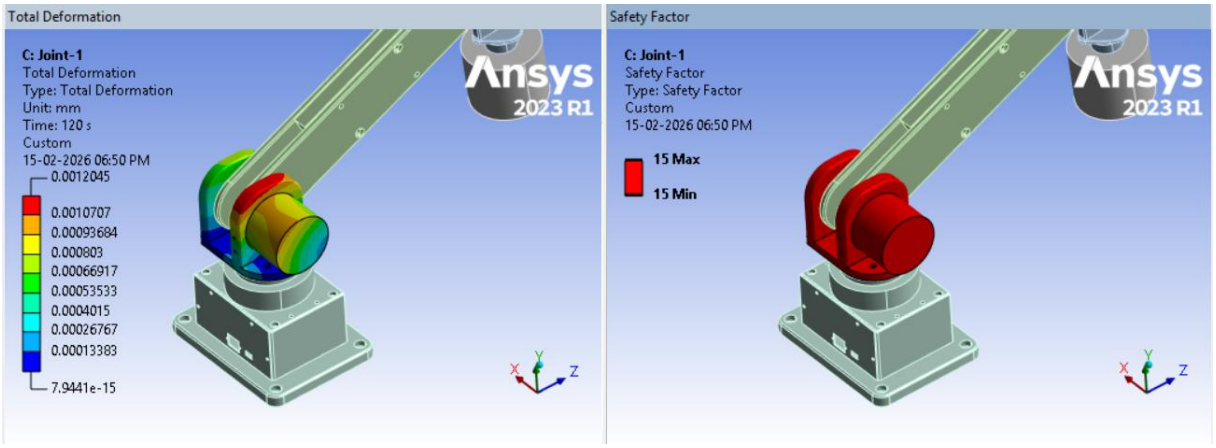


Figure-10: Total Deformation & FOS of joint body at motion

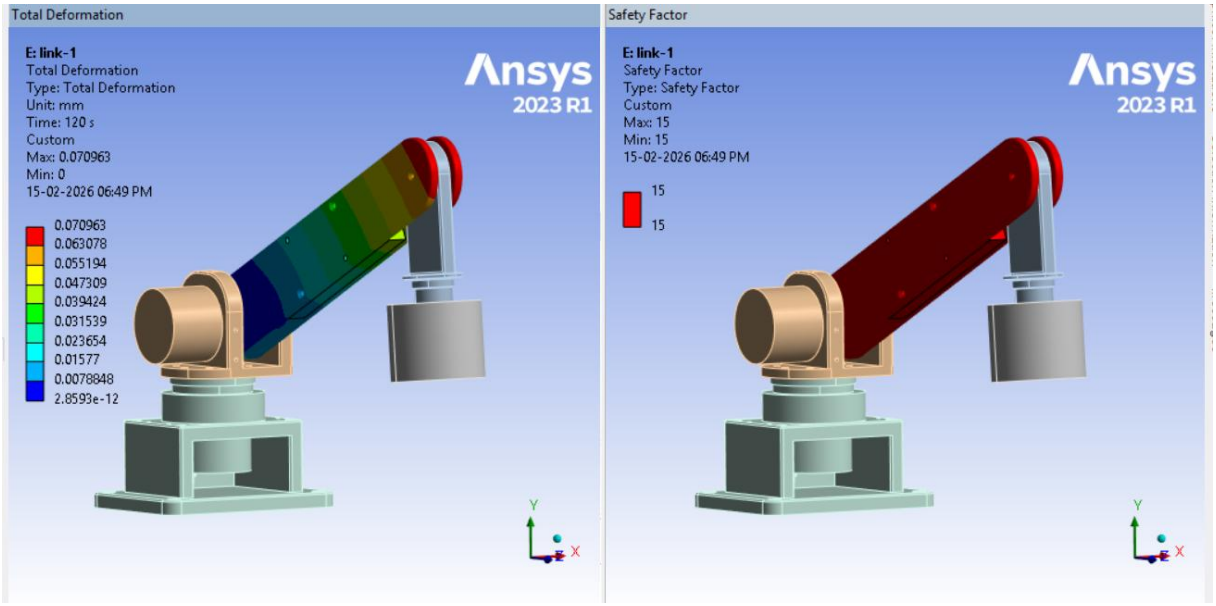


Figure-11: Total Deformation & FOS of link-1 body at motion

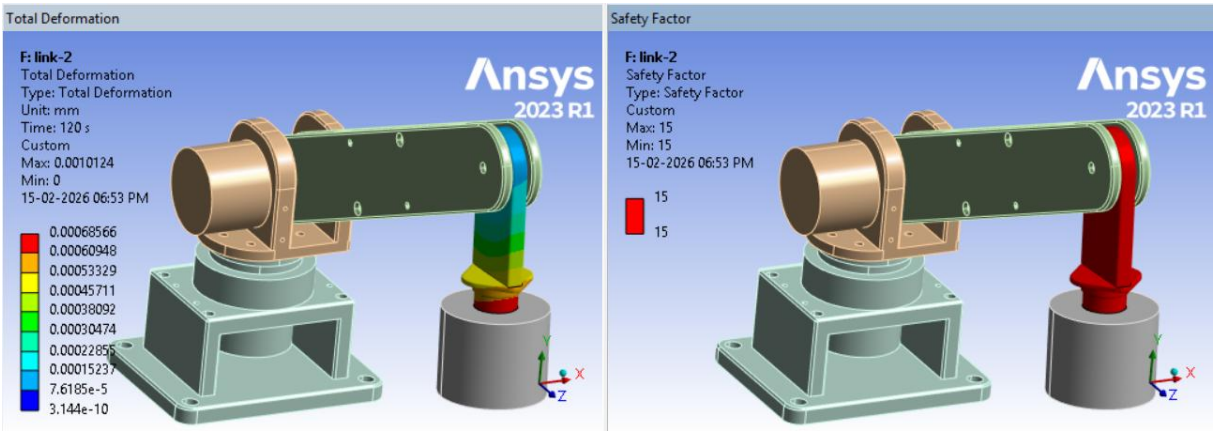


Figure-12: Total Deformation & FOS of link-2 body at motion

CONCLUSION:

From the Static & transient structural analysis, the minimum Factor of Safety (FOS) observed was 4.6, occurring at the joint region while holding the 5 kg payload after a 90° rotation.

Although this location experiences the highest stress during operation, the obtained FOS value is well above the acceptable design limit. Therefore, the structure is considered safe under the given loading and motion conditions.

The results confirm that:

- The joint assembly has sufficient strength to withstand dynamic loading.
- The selected material meets the structural requirements.
- The overall design provides adequate rigidity and strength.
- The robotic arm structure is suitable for 2-DOF pick-and-place operations.